

Ciguatera and scombroid fish poisoning in the United States.



BACKGROUND: Ciguatera and scombroid fish poisonings are common causes of fish-related foodborne illness in the United States; however, existing surveillance systems underestimate the overall human health impact. **OBJECTIVES:** This study aimed to describe existing data on ciguatera and scombroid fish poisonings from outbreak and poison control center reports and to estimate the overall number of ciguatera and scombroid fish-poisoning illnesses, hospitalizations, and deaths in the United States. **METHODS:** We analyzed outbreak data from the Foodborne Disease Outbreak Surveillance Systems (FDOSS) from 2000 to 2007 and poison control center call data from the National Poison Data System (NPDS) from 2005 to 2009 for reports of ciguatera and scombroid fish poisonings. Using a statistical model with many inputs, we adjusted the outbreak data for undercounting due to underreporting and underdiagnosis to generate estimates. Underreporting and underdiagnosis multipliers were derived from the poison control call data and the published literature. **RESULTS:** Annually, an average of 15 ciguatera and 28 scombroid fish-poisoning outbreaks, involving a total of 60 and 108 ill persons, respectively, were reported to FDOSS (2000-2007). NPDS reported an average of 173 exposure calls for ciguatoxin and 200 exposure calls for scombroid fish poisoning annually (2005-2009). After adjusting for undercounting, we estimated 15,910 (90% credible interval [CrI] 4140-37,408) ciguatera fish-poisoning illnesses annually, resulting in 343 (90% CrI 69-851) hospitalizations and three deaths (90% CrI 1-7). We estimated 35,142 (90% CrI: 10,496-78,128) scombroid fish-poisoning illnesses, resulting in 162 (90% CrI 0-558) hospitalizations and 0 deaths. **CONCLUSIONS:** Ciguatera and scombroid fish poisonings affect more Americans than reported in surveillance systems. Although additional data can improve these assessments, the estimated number of illnesses caused by seafood intoxication illuminates this public health problem. Efforts, including education, can reduce ciguatera and scombroid fish poisonings.